

Problem Set 1

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Problem 1

(i)

By Definition,

$$\begin{aligned} E[Y|X = x] &= \sum_y y \times p(y|X = x) \\ &= \sum_y y \times \frac{P(X = x, Y = y)}{P(X = x)} \end{aligned}$$

So,

$$E[Y|X = 1] = 2 * \frac{\frac{1}{36}}{\frac{1}{36}} = 2$$

(ii)

(iii)

(iv)

(v)

Problem 2

Problem 3

(i)

The log-likelihood for T i.i.d. observations is the following. We can prove the result.

(ii)

(iii)

The log-likelihood for T i.i.d. observations is the following. We can prove the result.

(iv)

Problem 4

(i)

The log-likelihood for T i.i.d. observations is the following. We can prove the result.

(ii)

Problem 5

(i)

The log-likelihood for T i.i.d. observations is the following. We can prove the result.

(ii)

Problem 6

(i)

The log-likelihood for T i.i.d. observations is the following. We can prove the result.

(ii)

(iii)

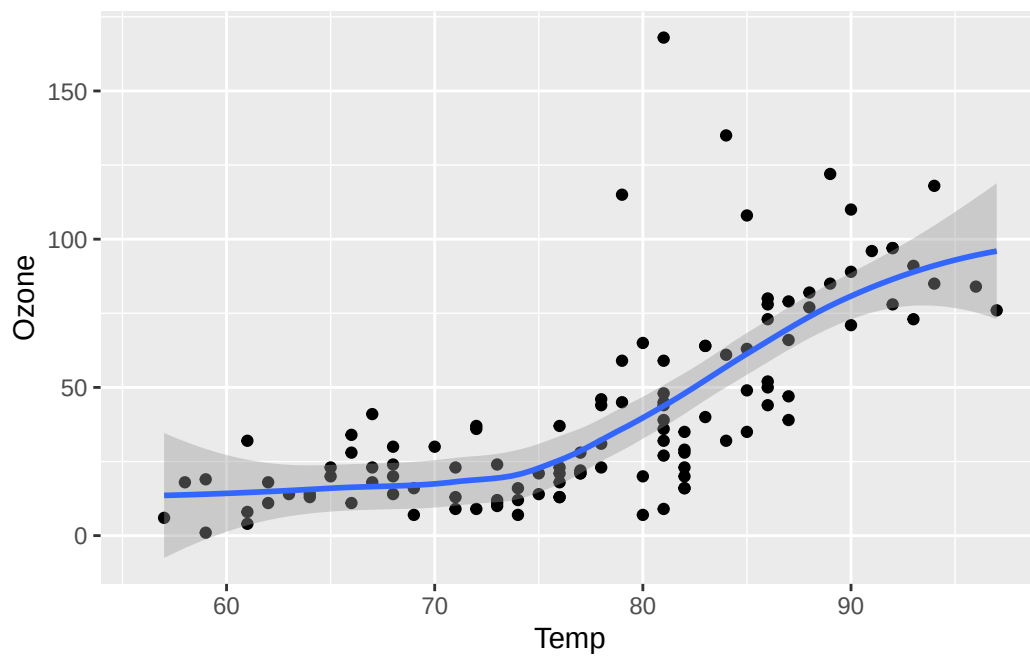


Figure 1: Temperature and ozone level.